

4(a)	E – haem / porphyrin (ring) ; I iron / iron-containing group A conjugated group / prosthetic group F – α helix ; R α helix G – tertiary structure ; A 3° structure H – primary structure ; A 1° structure J – quaternary structure / 4° structure / 2α and 2β globins ; A polypeptides for globins	5
4(b)(i)	98 (%) ;	1
4(b)(ii)	<i>four from:</i> - Bohr, shift / effect ; (at any pO_2 on the left of graph) percentage saturation of haemoglobin decreases / haemoglobin releases more oxygen (to muscle tissue) / increases dissociation of <u>oxyhaemoglobin</u> ; A any comparative data quote for any one value of pO_2 between 1 and 5 kPa (increase in carbon dioxide) <u>increases</u> production of carbonic acid ; <u>increase</u> in hydrogen ions (in red blood cells) ; - haemoglobin accepts hydrogen ions / formation of haemoglobinic acid (HHb) ; A haemoglobin has an affinity for hydrogen ions - decreases the <u>affinity</u> between haemoglobin and oxygen ;	4